



TEST PROCEDURE FOR MODEL 40623

Test Flow: Sensitivity and Polarity *

Noise

Weld

Leak Test

90% Offset/10% Noise (See Note)

Includes Diagnostic and Repair Guidelines

*This procedure includes several methods of measuring the sensitivity, refer to the Specification section to determine which applies to the filter being tested.

Note: If one detector fails in either test, then 100% test the remaining detectors for that test.

APPROVED BY / DATE	W	3460	11/14/24	<i>R7</i>	Revise procedure to call out AM502 Amplifier.
<i>Les Cooley 02-27-90</i>	V	3328	08/02/22	<i>EW</i>	Revised max sensitivity for detectors w/caloric FETs.
CHECKED BY / DATE	U	3318	07/14/22	<i>EW/LW</i>	Add spec for detectors w/caloric FETs.
<i>Lynn Wieland 08-23-88</i>	T	3240	06/09/21	<i>LW</i>	Add Note to Page 1.
PREPARED BY / DATE	S	3011	12/06/18	<i>LW</i>	Add diagnostic and repair guidelines.
<i>Dealmus Dixon 04-17-86</i>	REV	ECN	DATE	BY	DESCRIPTION

FORM # 067

40623 Sensitivity & Polarity

Equipment Required: Oscilloscope
Test Fixture 9000033
Test Fixture 9000157

Optical Bench
Digital Multimeter
Tektronix AM502 Amplifier

Set-Up: Frequency- 3 Hz (+/- .03 Hz) [Period @ 0.1 mSec scale: 0.3333 Sec, range: 0.3300 to 0.3367 Sec] (See Note 1)
Distance - 10 cm

Blackbody Temperature - 500 °K (* 227 °C +/- 1 °C for all set-ups)
(** 230 °C +/- 1 °C for set-up #5, 227°C +/- 1 °C for all other set-ups)

Aperture - .873" (full)

Connect Blade sync of chopper to oscilloscope (See Note 2).

Connect a "T" to the input on 9000157.

Connect output of 9000033 to one side of the "T" on 9000157.

Connect the other side of the "T" to a digital multimeter.

Connect output of 9000157 to the positive (+) input of the AM502 Amplifier: gain 100, LF = 0.1 Hz, HF = .1 k,
and +100 button out when necessary. (See Note 3)

Connect output of AM502 to CH1 of the oscilloscope.

Set oscilloscope as follows; CH1 to appropriate scale, DC coupled; Blade sync to 5 V/div, DC coupled and time
base to 100 mSec/div, Auto, AC coupled and trigger from Blade sync.

Procedure: Place detector in test fixture with tab to label.

Place a piece of cardboard between chopper and detector.

Observe offset reading on meter and wait until reading does not shift more than + / - 0.05 V.

Check against spec 1.

Remove cardboard and observe output and polarity of signal on oscilloscope, wait for maximum output reading.

Check against model specifications 2 and 3. (See Note 3)

Wait for maximum output reading. Wave form should be triangular [].

Specification: 1. Offset - min - 0.3 V
max - 1.2 V

2. Sensitivity - min - 28.6 mV w/-4 filter **
66.4 mV w/-5 filter *
6.18 mV w/-12 filter (Extra -4 filter and blackened tube) **
6.15 mV w/-23 filter **
28.1 mV w/-25 filter *
1.60 mV w/-79 filter (Extra -4 filter and blackened tube) **
2.47 mV w/-620 filter (Extra -4 filter and blackened tube) **
58.0 mV w/-621 filter *
5.23 mV w/-676 filter (Extra -4 filter and blackened tube) **
4.33 mV w/-700 filter (Extra -4 filter and blackened tube) **
2.43 mV w/-701 filter (Extra -4 filter and blackened tube) **
3.05 mV w/-702 filter (Extra -4 filter and blackened tube) **
1.79 mV w/-714 filter (Extra -4 filter and blackened tube) **
4.72 mV w/-716 filter (Extra -4 filter and blackened tube) **
2.79 mV w/-760 filter (Extra -4 filter and blackened tube) **

See pages 3 & 4 for other filters

3. Polarity should be positive for a positive change

Note: 1. For set-up # 2; Frequency- 3 Hz (+/- .09 Hz) [Period @ 0.1 mSec scale: 0.3333 Sec, range: 0.3233 to 0.3433 Sec].

2. Blade sync is CH2 on two channel oscilloscopes and CH4 on four channel oscilloscopes.

3. If the AM502 gain is used to measure the sensitivity, remember to divide the gain value of the AM502 out of the
sensitivity reading to obtain actual reading to compare to minimum sensitivity Specification.

40623 Sensitivity & Polarity

Equipment Required: Oscilloscope
Test Fixture 9000493
Test Fixture 9000157
Quartz Bulb Optical Bench 9000368
Digital Multimeter

Set-Up: Turn the quartz bulb bench ON and allow five minutes for it to stabilize
Frequency- 3 Hz (+/- .03 Hz) [Period @ 0.1 mSec scale: 0.3333 Sec, range: 0.3300 to 0.3367 Sec]
Connect a "T" to the input of test fixture 9000157.
Connect output of test fixture 9000493 to one side of the "T" and digital multimeter to the other side of the "T".
Connect output of test fixture 9000157 to CH1 of oscilloscope.
Connect blade sync of quartz bulb chopper to CH2 of oscilloscope.
Set switch on test fixture 9000493 to VM.
Set oscilloscope as follows; CH1 to appropriate scale, DC coupled; CH2 to 5 V/div, DC coupled and time base to 100 mSec/div, Auto, AC coupled and trigger off CH2.

Procedure: Place detector in test fixture with tab to tab label.
Place a piece of cardboard between chopper and detector.
Observe offset reading on meter and wait until reading does not shift more than $+/- 0.05$ V.
Check against specification 1.
Remove cardboard and observe output and polarity of signal on oscilloscope, wait for maximum output reading.
Wave form should be triangular .
Check against model specification 2.
Remove the detector.
Turn the quartz bulb bench OFF after all detectors are tested.

Specifications: 1. Offset - min - 0.3 V
max - 1.2 V
2. Sensitivity - min - 31.4 mV w/-723 filter (Extra -4 filter and blackened tube) See Note.
- 108 mV w/-749 filter
Polarity should be positive for a positive change

Note: Max specification for -723 w/Calogic FETs is 64 mV.

40623 (-113 Filter only) Sensitivity & Polarity

Equipment Required: Oscilloscope
Power Supply
Test Fixture 9000329
Test Fixture 9000157
Test Fixture 9000332 (100X Amplifier)
Test Fixture 9000355
Vertical Optical Bench
Three prong Male Banana Jacks with Shielded Cable
Nitrogen
Multimeter

Set-up: Frequency- 3 Hz (+/- .03 Hz) [Period @ 0.1 mSec scale: 0.3333 Sec, range: 0.3300 to 0.3367 Sec] (See Note1)
Distance - 10 cm
Blackbody Temperature – 500 °K (typically 214 °C or 215 °C)
Aperture - .873" (Full)
Connect a "T" to output of test fixture 9000355
Connect input of test fixture 9000157 to one side of the "T" and multimeter to other side of the "T".
Connect the nitrogen line to test fixture 9000329 and set flow rate to 0.2 scfh.
Connect the output of test fixture 9000157 to the input of test fixture 9000332.
Connect the output of test fixture 9000332 to the CH1 input of the oscilloscope.
Connect blade sync of chopper to CH2 of oscilloscope.
Set oscilloscope as follows; CH1 to appropriate scale, DC coupled; CH2 to 5 V/div , DC coupled and time base to 100 mSec/div, Auto, AC coupled and trigger off CH2..
Set the power supply to +/- 5 V dc (+/- 0.01 V dc) and turn OFF.
Connect power supply to test fixture 9000332 using the three prong male banana jacks with shielded cable observing polarity.
Check calibration of the set-up using the S451-12 reference detectors. (See Note 2)

Procedure: Remove test fixture 9000355 from chamber of test fixture 9000329 and place detector in test fixture with tab to label. (See Note 3)
Replace test fixture 9000355 onto chamber of test fixture 9000329.
Turn power supply ON.
Observe offset reading on meter and wait until reading does not shift more than + / - 0.05 V.
Check against specification 1.
Observe signal on oscilloscope. Keep in mind that the signal on the oscilloscope is 100 times the actual reading, so divide oscilloscope readings by 100 for the following steps.
Wait at least 1 min to 2 mins for maximum output reading.
Wave form should be triangular .
Observe output and polarity of signal on oscilloscope and check against model specification 2.
Remove detector.

Specifications: 1. Offset - min - 0.3 V
max - 1.2 V
2. Sensitivity - min - 1.93 mV with -113 under Nitrogen.
Polarity should be positive for a positive change

Notes: 1. For set-up # 2; Frequency- 3 Hz (+/- .09 Hz) [Period @ 0.1 mSec scale: 0.3333 Sec, range: 0.3233 to 0.3433 Sec]
2. Purge test fixture for 10 minutes at 1 scfh or 2 minutes at 5 scfh before testing detectors. Then set nitrogen flow rate to 0.2 scfh. The reference detectors should be measured twice each day, once in the morning and once in the afternoon. Keep in mind that the signal on the oscilloscope is 100 times the actual reading, so divide oscilloscope readings by 100. **Record** the sensitivity of the reference detectors for 10 test cycles, forward info to Eng Dept.
3. Test fixture 9000329 attaches directly to the face of the blackbody. Remove the blackbody's face and attach test fixture 9000329, using all four bolts. The test fixture consists of a box with an enclosed chopper which attaches

to the blackbody, a chamber with a -4 filter mounted on one end. Push test fixture 9000355 onto the tube and lower it to remove from the chamber.

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40623 Noise

Equipment Required: Multimeter
Power Supply (+/- 5 V dc)
Test Fixture 9000232
Test Fixture 9000233 with vacuum connection
Test Fixture 9000268 (Switch box)
Test Fixture 9000234 (Cable Assembly)
Test Fixture 9000272 (Rectifier Hold Circuit)
Test Fixture 9000433 (Green box)

Set-up: Set power supply to +/- 5 V dc (+/- 0.05 V dc) and turn OFF.
Connect the +/- 5 V dc power supply to test fixture 9000232, the rectifier hold circuit 9000272, and the cable assembly 9000234.
Connect the cable assembly to the 1-25 and 26-50 inputs of test fixture 9000233. Make sure the connector with the power supply connected to it goes to the 26-50 input. (See Note 1)
Connect the other side of the cable assembly to the switch box. Connect output of switch box to the input of test fixture 9000232.
Connect ground of switch box to ground of power supply.
Connect the output of test fixture 9000232 to the input of the rectifier hold circuit.
Connect the output of the rectifier hold circuit to the input of the multimeter.
Set multimeter to the DC volts, 200 mV scale.
Open test fixture 9000233 and set top aside.

Procedure: Place detectors into sockets with tabs pointing to the left.
Place lid onto test fixture 9000233, lock in place and pull vacuum.
Turn power supply ON.
Let detectors stand for five minutes for detectors to stabilize. (See Note 2)
Start testing with both switches on amplifier box on 1. Allow 15 to 20 seconds for detectors to settle.
Press and hold the Reset button on the rectifier hold circuit until digital multimeter reads less 1.0 mV.
Release Reset button and allow a minimum of 60 seconds for multimeter to settle.
Observe reading on multimeter and check against model specification. Record the noise levels of all detectors failing specifications. (Use numbers on data sheet to match numbers of both switches.)
Turn the column switch to 2, (this would read 1, 2). Allow 15 to 20 seconds and press the reset button. Check detectors as above and continue turning column switch to right and check each detector. (See Note 3)
When column switch reaches 10 and detector has been tested, turn row switch to 2. Then test detector, turn column switch to left and continue checking and turning column switch to left until it reaches 1, then turn row switch to 3. Continue this sequence until all detectors are tested.
Remove connector assembly from the 1-25 and 26-50 inputs of test fixture 9000233. Then connect it to the 51-75 and 76-100 inputs of test fixture 9000233. Make sure the connector with the power supply connected to it goes to the 76-100 input. Then repeat above steps.
Release vacuum, turn power supply OFF and unload and separate detectors. Acceptable detectors are placed on original tray. Place rejects on appropriate tray and record failure mode on data sheet in matching position. (See Note 4)

Specification: Noise level must be between 10.0 mV and 37.9 mV (under vacuum)

Note: 1. The lower half (51-100) may be measured at the same time by using an additional multimeter, 9000433, test fixtures 9000232, 9000234, 9000268 and 9000272. Connect the cable assembly to the 51-75 and 76-100 inputs of test fixture 9000233. Make sure the connector with the power leads is connected to the 76-100 input. Connect power leads of cable assembly to 9000433 (Green box). Connect test fixtures 9000232, 9000234, 9000268 and multimeter as above.
2. If after testing it is noticed that a longer stabilization time is required, contact Engineering.
3. The column switch is the knob which has a scale of 1 to 10, and the row switch is the knob which has a scale of 1 to 5.
4. Disposal of rejects: Preseal rejects are placed on a tray by model number and failure mode is recorded on data

sheet. Postseal rejects are placed in a storage box for retest in the future.

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40623 Offset Check

Equipment Required: Power Supply (+ 8 V dc)
Test Fixture 9000054 with 100 k Ω Rs (Test Box)
Test Fixture 9000056 (Amplifier) or 9000268 (Switch Box)
Digital Multimeter

Set-Up: Set the power supply to + 8 V (+/- 0.08 V dc) and turn OFF.
Connect power supply to amplifier and test box. [Apply ground to amplifier (or switch box) and 8 V to test box.]
Connect V in of amplifier to multimeter and set meter to 20 V dc setting.
Connect the amplifier (or switch box) to the left side of test box.

Procedure: Place detectors into sockets with tabs pointing to the left.
Close top of test box and lock in place.
Turn power supply ON.
An additional test set can be measured by repeating the above steps.
Let detectors stand for ten to fifteen minutes, if needed.
Set switch on amplifier to final. (If used)
Start testing with both switches on amplifier (or switch box) on 1.
Check offset on multimeter. If reading fails specification, record offset on back of history tag.
(Ex: Row 1 part 3 would be written as 1-3.)
Turn the column switch to 2, (this would read 1, 2). Check detector and continue turning column switch to right and check each detector. (See Note 1)
When column switch reaches 10 and detector has been tested, turn row switch to 2. Then test detector, turn Column switch to left and continue checking and turning column switch to left until it reaches 1, then turn row switch to 3.
Continue this sequence until all detectors are tested.
Remove connector from left side of test box and place on right side of test box. Then repeat above steps.
Turn power supply OFF.
Unload and separate detectors. Acceptable detectors are placed on original tray. Place rejects on appropriate tray and record failure mode on data sheet in matching position.

Specification: Offset - min - 0.3 V
max - 1.2 V

Note: 1. The column switch is the knob which has a scale of 1 to 10, and the row switch is the knob which has a scale of 1 to 5.

40623 Diagnostic and Repair Guidelines

Visually inspect failing detectors. If no visual defect is found as cause of failure, repair according to the following guidelines:

FAILURE MODE - Low offset, High offset, Shorted or Dead (LO, HO, SH, D)**REPAIR** - Replace FET.

Retest.

If detector fails again for same reason, carefully inspect for shorts. If short is found, repair appropriately. If no short is found, contact Engineering for repair instructions.

FAILURE MODE - Noisy (N)**REPAIR** - Replace crystal.

Retest.

If detector fails for high noise again, replace crystal.

Retest.

If detector fails for high noise again, scrap.

FAILURE MODE - Good offset w/no signal (GO/D)**REPAIR** - Replace crystal.

If detector fails again for good offset w/no signal, carefully inspect for shorts. If short is found, repair appropriately. If no short is found, scrap detector.

FAILURE MODE - Wrong wave shape (WW - not triangular as shown () or Wrong polarity (RP)**REPAIR** - Replace crystal.

If detector fails again for the same reason, take scope photo and send to Engineering for repair directions.

FAILURE MODE - Erratic**REPAIR** - Replace crystal.

Retest.

If detector fails again for erratic, take scope photo and send to Engineering for repair directions.

FAILURE MODE - Low sensitivity (LS) with the correct waveshape**REPAIR** - Replace crystal.

Retest.

If detector fails again for same reason, contact Engineering for repair directions.

FAILURE MODE - Any failure mode not listed above.**REPAIR** - Contact Engineering for repair directions.